

Scenario B, revised: sanitized HE, replace-one DP, privacy ledger

honest-but-curious parties · orange = new or changed vs. the draft protocol · numbered badges = risk fixed (key at bottom)

Researcher R
holds X_R , ID_R ; owns sk_{BFV} , sk_{CKKS}

Organization O
holds X_O , y , ID_O ; keeps the privacy ledger

Phase 0 – Setup and commitments (only public constants cross; no data-derived statistic leaves either party)

2 standardize X_R with R 's own constants (kept local)
clip rows: $\|x_{R,i}\| \leq B_R$ (intercept included)
KeyGen: BFV (PSI), CKKS (aggregation), 128-bit,
with modulus headroom for flooding

pk, evk, B_R, d_R

$\sigma, B_O, B_y, d_O, n_O,$
public D, α , hash key K

2 standardize with *public* constants; clip to B_O, B_y
3 $\Delta_{\text{rep}} \leftarrow$ replace-one sensitivity (see key)
4 $\sigma \leftarrow \text{AnalyticGauss}(\Delta_{\text{rep}}, \epsilon_{\text{msg}}, \delta)$
ledger: refuse unless $\rho_{\text{used}} + \rho_{\text{run}} \leq \rho_{\text{cap}}$
 $\rho_{\text{run}} = \Delta_{\text{rep}}^2 / 2\sigma^2 + \rho_{\text{err}}$ (zCDP)

Phase 1 – Private record linkage (BFV labeled PSI; R learns the match set; O learns only public sizes)

cuckoo-hash $H_K(ID_R)$ into bins
encrypt windowed powers y^{2^i} , $i < W$

$\text{Enc}(y^{2^i})$ per bin

5 secret permutation τ of rows; labels = τ -positions
simple-hash $H_K(ID_O)$; pad bins to *public* D, α
dummy root = $t - 1$ for the calibrated t

1 evaluate $P(y)$, $L(y)$ per partition (ct $\times$ pt)
2 masks $r_1, r_2 \xleftarrow{\$} \mathbb{Z}_t^*$ from a CSPRNG
3 $s = r_1 P(y)$, $q = r_2 P(y) + L(y)$
4 $s, q \leftarrow \text{Sanitize}_{\text{BFV}}(s), \text{Sanitize}_{\text{BFV}}(q)$

sanitized $\text{Enc}(s), \text{Enc}(q)$

decrypt: $I = \{k : s_k = 0\}$ and $k \mapsto j = q_k$
(j is a uniformly random τ -position)
build $b \in \{0, 1\}^{n_O}$ and $\tilde{X}_R$ in τ -order
 $A = \tilde{X}_R^\top \tilde{X}_R$ (exact, local, never sent)

Phase 2 – Encrypted aggregation and DP noise (CKKS, depth 1)

$\text{Enc}(b), \text{Enc}(\tilde{X}_R)$
(column chunks $\leq$ slots)

1 $\text{Enc}(B), \text{Enc}(C), \text{Enc}(c_R), \text{Enc}(c_O), \text{Enc}(y^\top y)$
2 inner products; the total sits in *every* slot
3 $z \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ per entry (symmetric on B), CSPRNG
4 discrete Gaussian, *same value in all slots*
 $ct \leftarrow \text{Sanitize}_{\text{CKKS}}(ct \boxplus \text{Enc}_{pk}(z))$
the fresh Enc re-randomizes c_1
ledger: $\rho_{\text{used}} += \rho_{\text{run}}$ before sending

sanitized $\text{Enc}(\tilde{B}), \text{Enc}(\tilde{C}),$
 $\text{Enc}(\tilde{c}_R), \text{Enc}(\tilde{c}_O), \text{Enc}(y^\top y)$

Phase 3 – Solve at R (post-processing only; no further privacy cost)

6 fix λ, Ψ from public (σ, p, n, ℓ) , *never* from $\tilde{G}$
 $\lambda = \max\{0, \rho^* 2\sigma\sqrt{p} - n\ell\}$, $\Psi = I$
decrypt slot 0; $\tilde{G} = \begin{bmatrix} A & \tilde{C} \\ \tilde{C}^\top & \tilde{B} \end{bmatrix}$, $\tilde{c} = \begin{bmatrix} \tilde{c}_R \\ \tilde{c}_O \end{bmatrix}$
 $\tilde{\beta}_\lambda = (\tilde{G} + \lambda\Psi)^{-1} \tilde{c}$
 $\hat{\beta}_{\text{bc}} = \tilde{\beta}_\lambda - \sigma^2 M(\tilde{P}_\lambda) \tilde{\beta}_\lambda$ (Thm 2, given $\hat{\rho} \geq 1$)

Released by R : $\hat{\beta}_{\text{bc}}$, with $\hat{\rho}$, the ridge shrinkage
 $-\lambda \tilde{P}_\lambda \Psi \tilde{\beta}_\lambda$, and RSS (floored at 0):
 $\tilde{y}^\top \tilde{y} - 2\tilde{c}^\top \tilde{\beta}_\lambda + \tilde{\beta}_\lambda^\top \tilde{G} \tilde{\beta}_\lambda$
(ϵ, δ)-DP for O 's matched individuals (x_O, y) ;
 O 's unmatched rows enter nothing R can see

Sanitize_{BFV}	ModSwitch ₀ (ct) $\boxplus$ Enc _{pk} (0) $\boxplus$ (e _{fl} , 0), $\ e_{\text{fl}}\ \approx 2^\kappa B_{\text{err}}$ with $N2^{-\kappa} \leq 2^{-40}$: statistical flooding (BFV decrypts exactly, so the extra error costs only modulus bits).		
Sanitize_{CKKS}	ModSwitch ₀ (ct) $\boxplus$ (e _{fl} , 0), $e_{\text{fl}} \sim \mathcal{D}_{\mathbb{Z}^N, \varsigma}$, $\varsigma = S_{\text{err}}/\sqrt{2\rho_{\text{err}}}$, so the leftover error channel is ρ_{err} -zCDP per record. DP-calibrated flooding needs a few bits above the error rather than 2^{40} (which would swamp CKKS precision); its cost ρ_{err} enters the ledger. The DP noise z must be one value shared by all slots: independent per-slot noise could be averaged away.		
Sensitivity	$\Delta_{\text{rep}} = \sup \ f(D) - f(D')\ _2$ over D, D' differing in one matched (x_O, y) with x_R fixed; $\Delta_{(6)} \leq \Delta_{\text{rep}} \leq 2\Delta_{(6)}$ (closed form in design.md). Keeping $y^\top y$ costs nothing under replace-one.		
Risks fixed	1 circuit privacy of O's replies (critical)	2 calibration leak, committed bug (crit./high)	3 replace-one sensitivity (high)
	4 no privacy ledger across runs (high)	5 row index, PSI metadata D, α (high, partial)	6 λ chosen from the noise (validity)
	7 floating-point noise, weak RNG (medium)		
Still revealed	by design (Def. 4): which of R 's cohort appear in O 's register, and n . Transport: mutually authenticated TLS; structured serialization, no pickle.		